

Mitigating Non-IID Challenges in Federated Learning: A Synthetic Data Augmentation Approach

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Abstract—Federated learning (FL) enables collaborative model training across hospitals, clinical centers, and medical devices while preserving data privacy. However, heterogeneous data characteristics—including class imbalance, limited annotations, and non-IID distributions—lead to substantial communication overhead, thereby restricting the scalability and efficiency of federated healthcare systems. To tackle the aforementioned challenges, this article introduces a federated learning algorithm that incorporates synthetic data augmentation within federated medical datasets to improve diagnostic accuracy and reduce communication overhead in distributed healthcare environments. By embedding synthetic data generation into the FL process, the proposed algorithm mitigates data heterogeneity, enhances model generalization, and reduces the number of communication rounds required for convergence under weakly supervised and imbalanced data conditions. The proposed algorithm is evaluated on two biomedical datasets—heart disease and diabetes—and demonstrates consistently superior performance over standard FL baselines in horizontal non-IID settings.

Index Terms—Federated learning, privacy, non-IID data, synthetic data, accuracy, data skew, attribute skew, label skew.

I. INTRODUCTION

In today's data-driven environment, a key challenge is balancing the maximization of data utility with the protection of user privacy and data security, particularly in domains involving sensitive information, such as medical and healthcare settings. In these contexts, data are often distributed across multiple sites (e.g., hospitals, specialist clinics, and patient devices), making it difficult to share datasets for training machine learning (ML) models. To address this, FL has emerged as a promising approach, enabling collaborative training of ML models across distributed, decentralized data sources without direct data exchange, thereby improving accuracy and performance while preserving privacy [1]–[3].

Unlike traditional centralized learning, FL allows participants to train a shared model locally and only share model updates with a central server [4]. This capability is particularly critical in healthcare, where patient data privacy is

a primary concern. However, medical records are inherently heterogeneous, with considerable variation in data distribution between institutions. FL can be categorized based on data partitioning: in horizontal FL, data are partitioned by samples, with institutions holding records of different patients but the same features; in vertical FL, data are partitioned by features, with institutions holding different attributes for the same patients; and federated transfer learning combines elements of both.

Despite these advantages, FL faces significant challenges when applied to non-independent and identically distributed (non-IID) data, which are prevalent in real-world medical datasets [5]–[7]. Such non-IID characteristics arise from differences in patient demographics, medical practices, and data collection methods, leading to heterogeneous data distributions that hinder model convergence and degrade predictive performance [8]. In practice, non-IID datasets consist of samples from different probability distributions that are not statistically independent. This heterogeneity can cause training drift and lead to performance degradation due to imbalances in feature and label distributions [5], [9], [10]. Therefore, effective model aggregation strategies and the stabilization of local training are essential for maintaining both accuracy and computational efficiency [11]. **The primary focus of this work is improving convergence and communication efficiency under non-IID conditions, while privacy considerations are discussed at both the system and threat-model levels.**

In this work, we present a novel protocol, FL_nonIID_Synth, which uses a tabular synthetic data generation method to support scalable, efficient, and accurate FL in healthcare, leveraging a trusted server. The key technical contribution is a synthetic data augmentation strategy specifically designed to address unbalanced, horizontally non-IID medical datasets, thereby improving both accuracy and communication efficiency. By generating initial model parameters from synthetic data tailored to the medical domain, our method mitigates the negative effects of data imbalance. The protocol introduces two distinct algorithmic cases, each optimized for different learning objectives. Experimental results show that our approach not only improves data balance but also reduces computational overhead by approximately 3 to 10 times compared to state-of-the-art methods while achieving greater predictive precision. This dual focus on computational efficiency and model precision positions FL_nonIID_Synth as a practical and robust solution for real-world healthcare applications where both resource constraints and model reliability are

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critical.

This article makes the following key contributions:

- **Mitigation of Non-IID data challenges** . We address attribute, label, and data-source skew in FL using a synthetic data augmentation strategy that creates balanced datasets. This improves the model's convergence and accuracy, reducing the mean squared error (MSE) to 5.2776 for the Diabetes dataset and 0.8504 for the Heart Disease dataset, outperforming conventional FL models.
- **Novel Synthetic Data Augmentation Algorithm** We introduce *FL_nonIID_Synth*, an FL algorithm that integrates synthetic data generation to overcome non-IID data limitations. The method reduces communication time by 3-10 times and achieves higher precision than baseline approaches on both the Diabetes and Heart Disease datasets.
- **Privacy-Preserving FL for Medical Data** We use synthetic data in place of raw patient records to ensure privacy while enabling collaborative training on sensitive datasets, such as Diabetes and Heart Disease, across distributed healthcare facilities.

The remainder of this paper is organized as follows. Section II discusses the challenges associated with non-IID healthcare data, including attribute, label, and data-source skew, as well as the corresponding threat model. Section III presents the proposed FL nonIID Synth methodology and synthetic data generation strategies. Section IV describes the implementation details, experimental setup, and performance evaluation. Section V reviews related work and compares the proposed approach with existing FL methodologies. Finally, Section VI concludes the paper and outlines future research directions.

II. THE COMPLEXITIES OF NON-IID HEALTHCARE DATA

Non-IID data are common in healthcare due to variations in patient demographics, medical practices, and data collection methods. Such heterogeneity can degrade the generalization and reliability of machine learning models. This section outlines three key types of skew in real-world medical datasets and summarizes potential mitigation strategies.

A. Skews in Real-World Datasets and Solutions

Attribute Skew: Occurs when datasets have different feature sets or representations.

Heart Disease Dataset: Contains 76 attributes from four sources, but only 14 common attributes are used to minimize skew.

Diabetes Dataset: Horizontally partitioned across hospitals with the same 41 attributes, reducing attribute skew.

Solutions: Select a common subset of attributes, standardize formats across sources, and use imputation or augmentation to handle missing features.

Label Skew: Arises when the distribution of target classes varies across datasets.

Heart Disease: Class imbalance between the presence and absence of disease between sources.

Diabetes: Variation in "times in hospital" between hospitals.

Solutions: Apply stratified sampling, synthetic data augmentation, weighted loss functions, or re-/oversampling to balance labels. Our work uses synthetic data generation to balance labels.

Data Source Skew: Results from demographic, procedural, and prevalence differences between data sources.

Present in both Heart Disease and Diabetes datasets despite common feature sets.

Solutions: Normalize and standardize features, perform feature engineering, use domain adaptation, apply data augmentation from underrepresented sources, and stratify sampling to ensure proportional representation.

B. Threat Model

Our healthcare FL approach's threat model includes several important components, particularly when dealing with non-IID data. Data privacy concerns are critical, as adversaries may attempt to access sensitive healthcare information and compromise patient confidentiality. Strict adherence to privacy regulations and the use of anonymization techniques are essential to mitigate this risk. Additionally, model accuracy may be affected by data skew caused by imbalances across healthcare facilities. Synthetic data generation is employed to address this issue by balancing datasets and improving model convergence.

Furthermore, risks arise during the transmission of data between medical clients and the central server, making secure communication protocols necessary to ensure data protection. Another important aspect is model security, which includes potential vulnerabilities during model updates that could lead to privacy breaches. To mitigate these risks, privacy-preserving strategies are incorporated into the FL process. While the proposed framework enhances privacy by avoiding direct sharing of raw patient data, the use of synthetic data introduces additional considerations. Although synthetic data does not directly expose original records, it may still retain statistical patterns that could be exploited by adversaries through attacks such as membership inference or model inversion. Therefore, it is important to ensure that the synthetic data generation process preserves privacy while maintaining utility. **It is important to note that Synthpop does not inherently provide formal differential privacy guarantees. Although synthetic data reduces direct exposure of raw patient records, there remains a possibility that generated samples may preserve identifiable statistical patterns from the original data. Consequently, attacks such as membership inference or model inversion may still pose potential risks under certain adversarial settings.**

Overall, the presented threat model outlines the key risks and high-level mitigation strategies to ensure the confidentiality, integrity, and security of the healthcare FL system. However, a more comprehensive analysis of these threats and the incorporation of advanced privacy-preserving techniques remain important directions for future work. In particular, future work will investigate the integration of formal privacy-preserving mechanisms, such as differential privacy, into the synthetic data generation process to provide quantifiable privacy guarantees while maintaining model utility.

III. METHODOLOGY

The proposed methodology consists of three main components. First, we focus on synthetic data generation using methods such as DataSynthes, SDV, and Synthpop. These techniques are applied to horizontally partitioned datasets collected from different hospitals, specifically, datasets on diabetes and heart disease. The objective is to identify the synthetic data generation method that delivers the best performance across these datasets.

Once the most effective method is identified, we apply our FL algorithm, *FL_nonIID_Synth*, to the selected synthetic datasets. This step enhances the robustness of machine learning models by addressing non-IID data challenges, including attribute, label, and data source skew.

To further address data imbalance, the proposed FL methodology incorporates a **data source balancing** strategy implemented in two distinct cases.

- A) **Client-side balancing:** Each participating client generates synthetic samples locally to balance its dataset before training.
- B) **Server-side balancing:** The central server generates synthetic samples to balance aggregated datasets after receiving model updates from clients.

Finally, we validate our approach through real-world applications using hospital datasets, demonstrating its practicality and effectiveness in sensitive healthcare settings.

A. Synthetic Data Generation methods and their comparison

Here, synthetic data generation plays a crucial role in addressing non-IID data challenges. We compared three synthetic data generation methods—DataSynthes, SDV, and synthpop—using real datasets, specifically diabetic and heart disease datasets. **The generated synthetic data are designed to approximate the statistical properties of the original datasets, preserving key feature distributions and correlations. Although synthetic samples do not exactly replicate the true data distribution, they provide a closer approximation to a balanced global distribution across clients. This helps mitigate the divergence caused by non-IID data while maintaining the underlying structure of the original data.** Initially, we generated synthetic data from these real datasets to facilitate initial model generation, aiming to reduce communication complexity by minimizing the number of iterations required in the FL process. This step is vital because it streamlines the model training process, making it more efficient and less resource-intensive.

Secondly, we used synthetic data to augment the datasets. There are two cases for this augmentation. In the first case, we generate synthetic data to balance each client's dataset, thereby minimizing data imbalance. In the second case, a percentage of the combined synthetic dataset is sent to all clients, helping minimize attribute and label skew. By implementing these strategies, we effectively bring about explicit segmentation techniques, to ensure that the data accurately reflect the underlying variances in patient demographics, medical conditions, and treatment patterns across various geographic regions, to effectively address the multifaceted challenges posed by

data heterogeneity within healthcare systems, particularly in scenarios where the dataset exhibits non-IID characteristics across different data silos, as shown in Figure 1 and Figure 2.

We utilized datasets collected from multiple healthcare facilities located in different geographical regions to evaluate the proposed framework. The datasets capture variations in patient demographics, medical conditions, and treatment practices across institutions, thereby reflecting realistic non-IID characteristics commonly observed in healthcare environments. To protect patient confidentiality, all data were handled in accordance with privacy regulations and appropriate anonymization procedures. Using these heterogeneous healthcare datasets enables the development and evaluation of machine learning models that better reflect real-world variations in distributed medical data.

Our approach is designed to significantly enhance the accuracy of healthcare FL systems when dealing with non-IID data with a skew in quantity. These data-format inaccuracies have a devastating impact on *FedAvg*. Specifically, we address the challenge of varying data distributions from different geographical locations, which give rise to the aforementioned data skew. The FL process is initiated by disseminating a global model to multiple local devices. Subsequently, only the locally modified model weights are transmitted back to a central coordinating server, which is thoughtfully aggregated to refine and create an increasingly improved global model iteratively. This iterative process is executed over multiple rounds, with the global model continually improving based on the valuable contributions of each participating device or server, thus optimizing the accuracy of the FL framework in the healthcare domain. Figures 1 and Figure 2 explain the general methodology of the proposed algorithm. Starting from the main issue of reduced accuracy in FL systems when dealing with non-IID data, we address this problem in two cases: Figure 1 and Figure 2.

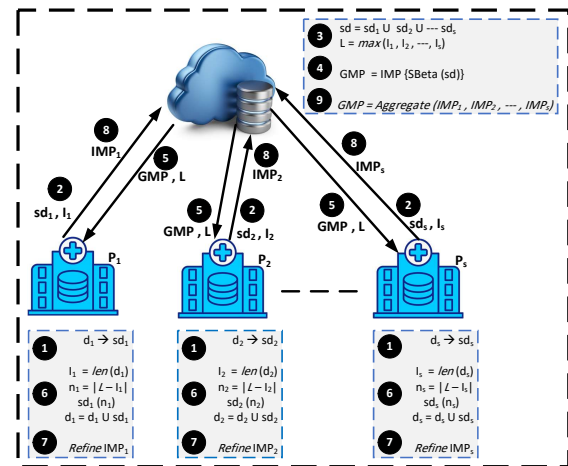


Fig. 1: Proposed FL methodology designed for non-IID distributions: Data source balancing from the client's side.

Case A: Investigating FL-based accuracy improvement based on data collected from different hospitals, which can have varying patient records. This data imbalance affects model

training and accuracy. Our solution involves generating synthetic data to balance the datasets, ensuring that each hospital's data is more evenly distributed. This helps the models converge faster and leads to better accuracy. The main steps are the following.

- 1) Initially each client generates synthetic data sd_i from their private data d_i , also calculate the number of records l_i .
- 2) sd_i and l_i are sent to the server.
- 3) The server finds the union of synthetic data SD from the received synthetic data from the client side. Also, evaluate the maximum number of records $L = \max(l_1, l_2, \dots, l_s)$.
- 4) Server finds the initial parameter $sBeta$ from SD and set $GMP = sBeta$. Here, $sBeta \in \mathbb{R}^d$ denotes the initial model parameter vector estimated from the synthetic dataset SD , obtained by training the base model (e.g., linear regression) on SD .
- 5) The server sends L and GMP to all clients.
- 6) Each client calculates the number of records $n_i = \lfloor L - l_i \rfloor$, selects randomly n_i number of synthetic data records $D_i = d_i \cup sd_i$.
- 7) Clients calculate their local model LMP_i .
- 8) Clients send the LMP_i to the server.
- 9) The server aggregates these LMP_i to get the global model parameter GMP . This GMP is sent to all clients, and the process repeats until the predefined termination condition is reached.

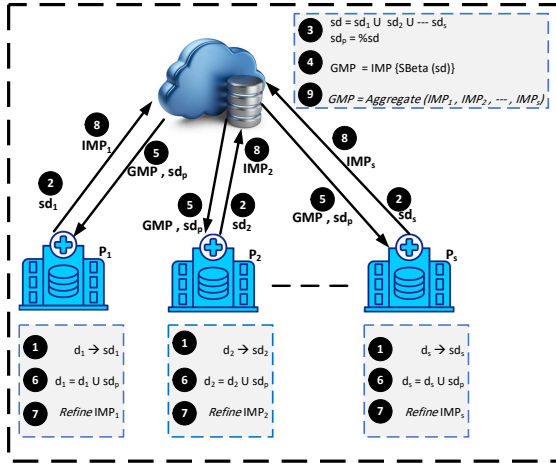


Fig. 2: Proposed FL methodology designed for non-IID distributions: Data source balancing from the server side.

Case B: Using clients to generate synthetic data and share it with a central server. The server then distributes a portion of the synthetic data back to all clients. This process reduces various types of data skew, including differences in data sources, attributes, labels, and timeframes, ultimately enhancing the accuracy of the FL system in the presence of non-IID data. Algorithm 2 explains this method. The steps are given below:

- 1) Initially each client generates synthetic data sd_i from their private data d_i
- 2) sd_i is sent to the server.
- 3) The server evaluates a percentage of data from the union of the datasets sd_i s. The server computes the union of

synthetic data (SD) from the synthetic data received from clients. Calculate $sd_p = \%SD$.

- 4) Find the initial parameter $sBeta$ from SD and set $GMP = sBeta$.
- 5) The server sends sd_p and GMP to all clients.
- 6) Each client combines $D_i = d_i \cup sd_p$.
- 7) Clients calculate their local model LMP_i .
- 8) Clients send the LMP_i to the server.
- 9) The server aggregates these LMP_i to get the global model parameter GMP . This GMP is sent to all clients, and the process repeats until the predefined termination condition is reached.

Algorithm 1 Model – Generation

for each round $t = 1, 2, \dots, T$ **do**

Step 1 - Each Client

Receives global model $GMP = w^{(t-1)}$ from server

Each client initializes local model $w_i^{(t,0)} \leftarrow w^{(t-1)}$

Each client's dataset d_i is split into batches of size B

Local model update initialized as $w_i^{(t,0)}$

for each local epoch $e = 1, 2, \dots, E$ **do**

for each local batch $b \in B$ **do**

$$w_i^{(t,e)} \leftarrow w_i^{(t,e-1)} - \eta \nabla L(w_i^{(t,e-1)}; b)$$

end for

end for

Send updated local model $w_i^{(t,E)}$ to server

Step 2 - Server

Receive $w_i^{(t,E)}$ from each client i

Perform aggregation:

$$w^{(t)} = \text{aggregation}(w_1^{(t,E)}, w_2^{(t,E)}, \dots, w_s^{(t,E)})$$

Set $GMP = w^{(t)}$

Server sends this GMP to all clients and proceeds to next round

end for

Stopping criterion: These steps are repeated until convergence, defined as $\|w^{(t)} - w^{(t-1)}\| < \epsilon$, or until a predefined maximum number of communication rounds T is reached.

The effectiveness of synthetic data augmentation in FL can be explained from an optimization perspective. In non-IID settings, local client updates diverge due to differences in data distributions, leading to slower convergence and instability in global aggregation. By augmenting local datasets with synthetic samples that approximate a more balanced distribution, the variance between local gradients is reduced. This results in more consistent local updates and improves the stability of the global optimization process. Consequently, the number of communication rounds required for convergence is reduced. Let g_i denote the local gradient at client i . In non-IID settings, $\mathbb{E}[g_i] \neq \mathbb{E}[g_j]$ for $i \neq j$. Synthetic data augmentation reduces this discrepancy by approximating a shared distribution, leading

to:

$$\text{Var}(g_i) \downarrow \Rightarrow \text{faster convergence.}$$

In Algorithm 1, $w_i^{(t,e)}$ denotes the local model at client i after epoch e in communication round t . After completing all local epochs, the final local model $w_i^{(t,E)}$ is sent to the server for aggregation. The aggregation step uses the final local models from all clients, i.e., $w_i^{(t,E)}$, to compute the updated global model $w^{(t)}$.

The novelty of our approach lies in using Synthpop to generate synthetic data to balance datasets and mitigate data skew in non-IID data scenarios. Unlike traditional methods, Synthpop enables the creation of realistic synthetic data that preserves the statistical properties of the original data, thereby improving model training and accuracy. Furthermore, our approach of distributing synthetic data to clients and iteratively refining the global model ensures that the FL system can effectively handle diverse and imbalanced datasets, thereby enhancing the overall accuracy and reliability of healthcare decision-making systems.

To demonstrate the effectiveness of our FL framework for non-IID datasets, we consider diabetes and heart disease datasets collected from different hospitals. The collected data involves quantity skewness in horizontally partitioned tabular data, where clients have different record counts, which falls under data skewness. Another form of data skewness is class imbalance, in which classes or outcomes are not evenly distributed across hospitals. In a binary classification task for disease prediction, one hospital's split may have many more positive instances than the other, creating an uneven class distribution. It is well known that this imbalance can hinder the model's generalization between hospitals. In addition, the collected data exhibit volume skewness, and the contribution of data to the partitions varies with hospital size or patient population. Generating more data in larger hospitals can result in uneven partition sizes. Skewness in data volume can affect both model training and evaluation.

IV. IMPLEMENTATION AND EXPERIMENTAL SETUP

A. Implementation.

We implement FL_non-IID_Synth using Python 3.8. Python was extended using the Scikit-learn package and the NumPy, pandas, and libraries for synthetic data generation. Our evaluation primarily uses two metrics: the accuracy of the global model on test data and the total communication required between the clients and the server.

B. DataSets.

1) *Heart Disease Dataset*: The provided dataset is from 1988 and comprises four distinct databases [12]: Cleveland, Hungary, Switzerland, and Long Beach V. The dataset contains 76 attributes, with the predicted attribute among them. However, all published experiments to date have used only a subset of 14 attributes. The dataset is horizontally partitioned; therefore, all clients share the same attribute set. The "target" field indicates the presence of heart disease in the patient. The variable is assigned integer values, where 0 represents the absence of disease and 1 represents the presence of disease.

2) *Diabetic Dataset*: The Diabetic dataset is found in the Cerner clinical database [13], a collection of data from various hospitals. The dataset is horizontally partitioned as their attribute space is common. The clinical dataset for diabetics comprises 9926 records and 41 features. It is divided among five clients. The dataset comprises 41 variables: hospital ID, race, gender, age, weight, admission type, glucose readings at different time points, insulin dosages, and additional variables. The variable 'times in hospital' denotes the number of days of hospitalization, ranging from 1 to 14 days. Clinical and demographic characteristics of Diabetic patients can be used to develop predictive models to estimate hospitalization duration.

The datasets collected are horizontally partitioned and share the same attributes but contain different records. The initial step is to identify the most efficient approach to generating synthetic data explicitly designed for tabular datasets. After conducting several tests on different methods, it was observed that Synthpop consistently generated synthetic datasets of superior quality compared to other statistical techniques. The experiments conducted in our study focused on analyzing heart disease and Diabetic data. These experiments demonstrated the effectiveness and appropriateness of using Synthpop to improve the accuracy and usefulness of synthetic data for these particular datasets.

C. Results and Discussion.

In this work, the communication cost refers to the total time required for model synchronization and data exchange between clients and the central server during the FL process. The reported total time includes local model training, model parameter transmission, synthetic data transmission, and server-side aggregation operations. In addition, the overhead associated with generating and distributing synthetic data is included in the reported execution time measurements.

The total communication time can be expressed as:

$$T_{total} = T_{train} + T_{model} + T_{synth} + T_{agg} \quad (1)$$

where T_{train} denotes local training time, T_{model} represents model parameter communication time, T_{synth} includes synthetic data generation and transmission overhead, and T_{agg} corresponds to server-side aggregation time.

TABLE I: Time taken (seconds) to generate 1 Million records of synthetic data by different synthetic data generation methods

Synthetic data generation methods	Diabetic data (39 attributes)	Heart Disease data (12 attributes)
Data synthesizer	0.658587	0.244847
SDV	306.9496	255.762898
Synthpop	23.756315	3.6463

Table I compares the time taken (in seconds) to generate 1 million synthetic data records using three different methods for diabetes data (39 attributes) and Heart Disease data (12 attributes). The Data Synthesizer is the fastest, with generation

times of approximately 0.659 seconds for Diabetic data and 0.245 seconds for Heart Disease data. In contrast, SDV (Synthetic Data Vault) is the slowest method, taking roughly 307 seconds for Diabetic data and 256 seconds for Heart Disease data, making it significantly slower than the Data Synthesizer. Synthpop falls in between, taking approximately 24 seconds for Diabetic data and 3.646 seconds for Heart disease data. These results highlight substantial time-efficiency differences between the methods, which should be considered alongside data quality when selecting a synthetic data generation approach for specific tasks. As the number of attributes increases, the time required to generate synthetic data for SDV and Synthpop increases. **The reported execution times therefore reflect the complete operational cost of the proposed framework, including the additional overhead introduced by synthetic data generation and communication.**

We implement FL non-IID Synth using Python and use linear regression as the machine learning model. The use of linear regression in this study is intended as a proof-of-concept to isolate the impact of the proposed FL nonIID Synth framework on convergence behavior and communication efficiency under non-IID conditions. The global model is initially set to 0. The FedAvg methodology was utilized to combine local models. The calculation of the global model uses an averaging technique. The algorithms' termination criteria are determined by either reaching the predefined maximum number of iterations or observing a minimal change in the mean square error (MSE). **In our experiments, convergence is assumed when the absolute difference between two consecutive MSE values is less than $\epsilon = 10^{-3}$, i.e.,**

$$|MSE^{(t)} - MSE^{(t-1)}| < \epsilon.$$

The value $\epsilon = 10^{-3}$ was selected to avoid premature stopping while ensuring that training terminates once the model improvement becomes negligible. The datasets were randomly partitioned into a training set (70% of the data) and a testing set (30% of the data). A regression model was developed using the training set and then applied to predict the outcome variable in the testing set.

The graph below shows experimental results comparing real data to synthetic data generated by the above approaches. It reveals how well synthetic data replicates real data. This graph shows how closely the synthetic data matches the real data, demonstrating how well the data creation algorithms capture the complex patterns and statistical aspects of the original dataset. Data analytics, model training, and privacy preservation depend on the reliability and integrity of synthetic data, which we can assess by plotting these data points. This graph aids data-driven decision-making by assessing the practicality and robustness of each synthetic data generation method that resembles real data. The experiments are conducted on both the Diabetic and heart disease datasets.

1) *Using Diabetic Data:* Figures 3(a), (b), and (c) visually compare the Correlation matrices for Datasynthesize, SDV, and Synthpop about Diabetic data. This comparison provides valuable insights into the quality and accuracy of the synthetic data generated by these methods. Correlation matrices are a valuable tool for evaluating the effectiveness of classification

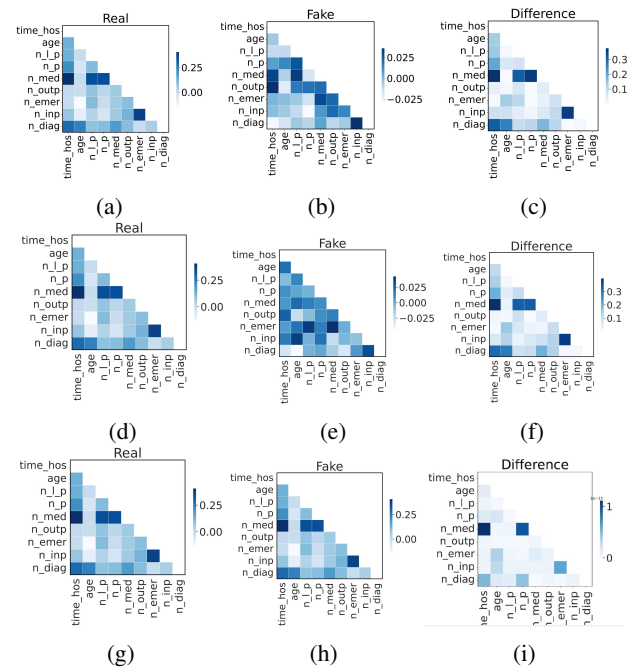


Fig. 3: Correlation matrix for Diabetic data



Fig. 4: First two components of PCA for Diabetic data

models. In Figure 3, they are used to gauge how different synthetic data generation methods accurately mimic the patterns observed in real data. The Synthpop method effectively generates synthetic Diabetic data that closely resembles the patterns and characteristics of real data. The Correlation matrices for Synthpop demonstrate that its classification results are nearly identical to those obtained with real data. The observations in Figures 4 and 5, which indicate that the synthetic data (fake data) generated by Synthpop are more accurate than those of other methods, carry significant implications for the quality and fidelity of the synthetic data produced by Synthpop.

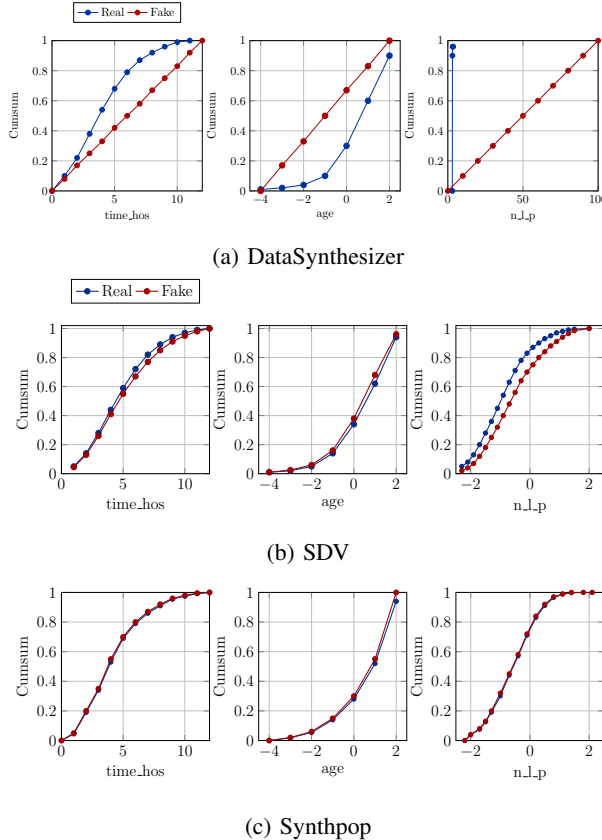


Fig. 5: Cumulative sum of few features in Diabetic data

2) *Using Heart Disease Data:* The described figures, including Figure 6 showing the correlation matrices for Heart Disease data, Figure 7 depicting the first two components of the principal component analysis and Figure 8 illustrating the cumulative sum of select features in Heart Disease data, collectively offer a comprehensive visual comparison of synthetic data generated by Datasynthesize, SDV, and Synthpop for Heart Disease data.

The figures appear to provide visual evidence of the effectiveness of the Synthpop method in accurately reproducing the key features and patterns observed in the actual data. The precision of synthetic data generation is highly valuable across a wide range of applications, including machine learning, data analysis, and privacy protection. Suppose the synthetic data generated by Synthpop exhibits a high degree of similarity to the real data. In that case, it implies that it can serve as a reliable alternative to real data in situations where the actual

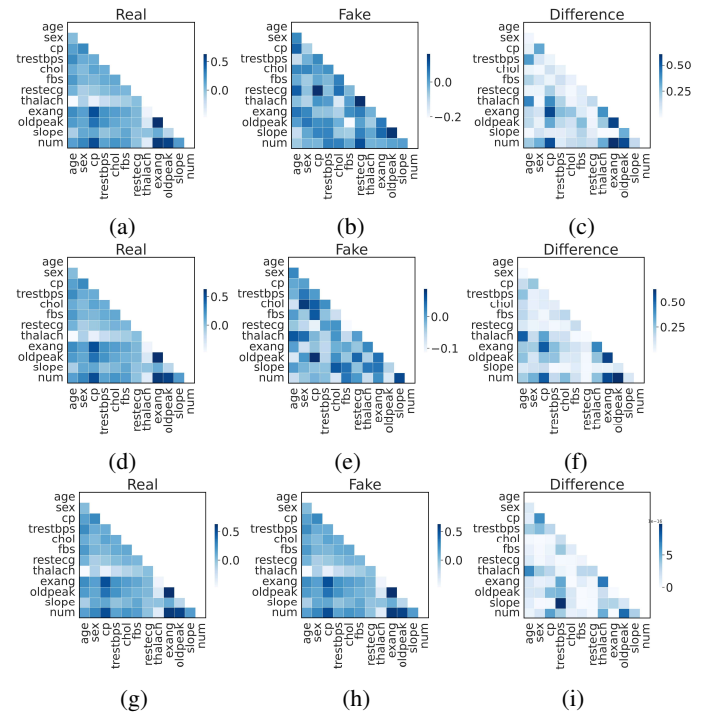


Fig. 6: Correlation matrix for Heart Disease data

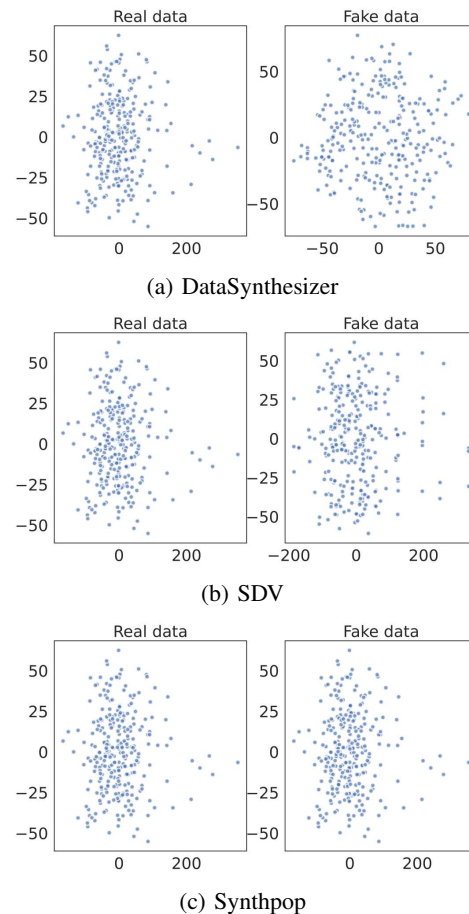


Fig. 7: First two components of PCA for Heart disease data

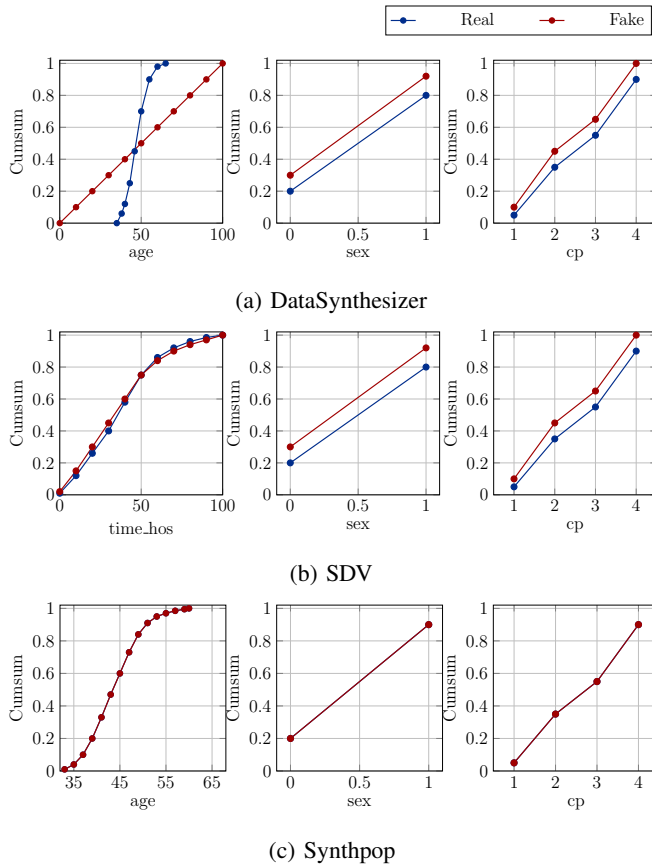


Fig. 8: Cumulative sum of few features in Heart disease data

data may be sensitive, limited, or difficult to obtain. Synthetic data is especially advantageous when privacy and data security are paramount. It enables analysis and model training without compromising result quality. The figures presented highlight the advantages of the Synthpop method as an effective tool for producing high-quality synthetic data. This data closely resembles real data, enhancing the accuracy and security of data-driven processes.

Choosing Synthpop as the synthetic data generation method is logical, supported by the positive outcomes and comparisons shown in the previous figures. The choice of using this method to generate synthetic data appears to be strong, as it can produce high-quality data that closely matches the patterns and features of the original dataset. The use of synthetic data can be highly beneficial in situations where privacy, security, or scarcity are concerns. It enables the generation of reliable data for analysis, modeling, or other applications. Selecting a synthetic data generation method that closely matches the characteristics of real data can significantly improve the quality and dependability of data-driven processes.

3) *Quantitative Evaluation of Synthetic Data Quality:* To complement the visual analysis presented in Figures 3–8, we additionally evaluate the quality of the generated synthetic data using a quantitative distribution similarity metric. Specifically, Kullback–Leibler (KL) divergence is used to measure the difference between the probability

TABLE II: KL divergence comparison between real and synthetic datasets

Method	Diabetes	Heart Disease
DataSynthesizer	0.184	0.162
SDV	0.097	0.089
Synthpop	0.041	0.036

distributions of the original and synthetic datasets. Lower KL divergence values indicate closer alignment between the generated synthetic data and the real data distribution.

The results in Table II quantitatively confirm the observations from the visual analysis. Synthpop achieves the lowest KL divergence values for both datasets, indicating that its generated synthetic samples more closely approximate the original data distributions compared to DataSynthesizer and SDV. This supports the selection of Synthpop as the synthetic data generation method used in the proposed FL framework.

4) *FL_{non-IID_Synth} Algorithm 2:* The approach described in Algorithm 2 Case A, where the data are horizontally partitioned, involves augmenting each client's dataset with synthetic data to achieve a balanced distribution of records across all clients, and is a viable technique for addressing non-IID bias in the context of FL. When data from different clients are not identically and independently distributed, with clients having varying amounts of data or data distributions, this can lead to biased and ineffective model training. Implementing a strategy that guarantees an equal distribution of records among clients mitigates the non-IID skew, leading to more equitable and well-balanced model updates in the context of FL. This methodology enhances the fairness and efficiency of the training process by enabling the model to learn from a diverse, unbiased subset of data from each client. As a result, the overall performance and equity of the FL system are improved.

In Table III, the Mean Square Error (MSE) values for two FL methods, "FL-Normal" and "FL_{non-IID_Synth}," are provided for both Diabetic and Heart Disease datasets. A lower MSE indicates better accuracy, representing a smaller difference between predicted and actual values. Comparing the MSE values, "FL_{non-IID_Synth}" outperforms "FL-Normal" regarding accuracy for Diabetic and Heart Disease data. "FL_{non-IID_Synth}" provides more accurate model predictions on both datasets, making it a better choice.

TABLE III: Mean Square Error (MSE) for Normal Federated Algorithm FL-Normal and Our Proposed Algorithm

Methods	MSE for Diabetic data	MSE for Heart Disease data
FL-Normal	5.3176	0.9947
FL _{non-IID_Synth}	5.2776	0.8504

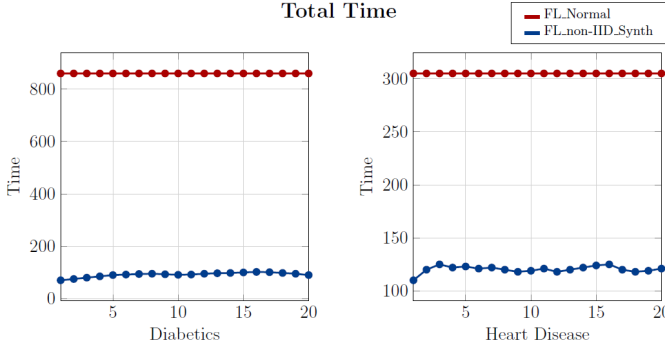


Fig. 9: The Total Time Taken for FL_{Normal} and $FL_{non-IID_Synth}$ for All the 20 Repetitions.

Algorithm 2 $FL_{non-IID_Synth}$ Algorithm

Initialization

Input

$C_1, C_2, \dots, C_S$: S clients with private data $d_1, d_2, \dots, d_s$ (horizontally partitioned). $w^{(0)}$: initial global model parameter (IMP), $w^{(t)}$: global model parameter at communication round t (GMP), η : learning rate, T : total iterations, B : minibatch size, ∇L : loss function.

Case A

Step 1 - Each Client

Each client generates a synthetic dataset ($sd_1, \dots, sd_s$) with their private data $d_1, \dots, d_s$, computes $l_i = \text{len}(d_i)$, and sends sd_i, l_i to the server.

Step 2 - Server

Union all sd_i to form Sd , compute $L = \max(l_1, \dots, l_s)$, initialize $sBeta$ from Sd , set $GMP = sBeta$, and send L, GMP to clients.

Step 3 - Each Client

Calculate $n_i = |L - l_i|$, randomly select n_i samples from sd_i , combine with d_i to form $D_i = d_i \cup sd_i$, call *Model - Generation*.

Case B

Step 1 - Each Client

Each client generates sd_i from d_i and sends it to the server.

Step 2 - Server

From $Sd = sd_1 \dots sd_s$, extract percentage $Sdp = \%Sd$, initialize $sBeta$ from Sd , set $GMP = sBeta$, send Sdp, GMP to clients.

Step 3 - Each Client

Combine Sdp with d_i to form $D_i = d_i \cup sdp$, call *Model - Generation*.

Figure 9 compares the total time taken for two FL scenarios, FL_{Normal} and $FL_{non-IID_Synth}$, over 20 repetitions for diabetic and heart disease data. In both cases, the FL_{Normal} scenario (red line) consistently takes signifi-

cantly longer. Conversely, the $FL_{non-IID_Synth}$ scenario (blue line) shows a much lower time.

TABLE IV: Total Time Taken by FL_{Normal} and $FL_{non-IID_Synth}$ (Avg) for Diabetic and Heart Disease Datasets

Datasets	Total Time (Seconds)		Total Number of Iterations	
	FL_{Normal}	$FL_{non-IID_Synth}$ (Avg)	FL_{Normal}	$FL_{non-IID_Synth}$ (Avg)
Diabetic data	865.209	79.984	145	13
Heart Disease data	311.768	119.658	76	28

Table IV compares the performance of the FL_{Normal} and $FL_{non-IID_Synth}$ algorithms on diabetic and heart disease datasets, focusing on total time (seconds) and the total number of iterations. The $FL_{non-IID_Synth}$ algorithm, which was averaged over 20 runs, significantly outperformed the FL_{Normal} algorithm in terms of both metrics. For the diabetic data, $FL_{non-IID_Synth}$ reduced the total time from 865.209 seconds to 79.984 seconds and decreased the number of iterations from 145 to 13. Similarly, for the heart disease data, $FL_{non-IID_Synth}$ lowered the total time from 311.768 seconds to 119.658 seconds and reduced the iterations from 76 to 28. These results indicate that the $FL_{non-IID_Synth}$ algorithm is more efficient, both in computation time and the number of iterations required, compared to the FL_{Normal} algorithm.

Figure 10 further illustrates the convergence behavior of the proposed FL non-IID Synth framework under non-IID conditions. Compared with the original non-IID data, the synthetic data augmented approach achieves faster convergence and lower MSE values across iterations. In particular, Figures 10(a)–10(d) show that synthetic data augmentation reduces the impact of data imbalance and improves training stability. However, some fluctuations remain visible, particularly in Figure 10(b). These variations can be attributed to residual heterogeneity in the client data distributions, differences in local dataset sizes, and the stochastic nature of synthetic data generation. Although synthetic data augmentation substantially reduces gradient divergence and improves convergence, small oscillations may still occur during local updates and global aggregation. Nevertheless, the proposed framework consistently converges more rapidly and achieves lower overall error than the standard FL approach.

5) *Contribution of Individual Components*: To analyze the contribution of each component in the proposed framework, we examine the impact of synthetic data generation, client-side balancing, and server-side distribution, as summarized in Table V.

The results indicate that each component contributes to performance improvement in different ways. Synthetic data initialization (Synth. Init) enhances the initial training phase by reducing data imbalance across clients. Client-side balancing further improves convergence by equalizing local datasets, leading to lower MSE values for both datasets. Server-side distribution improves consistency across clients by mitigating attribute and label skew. The

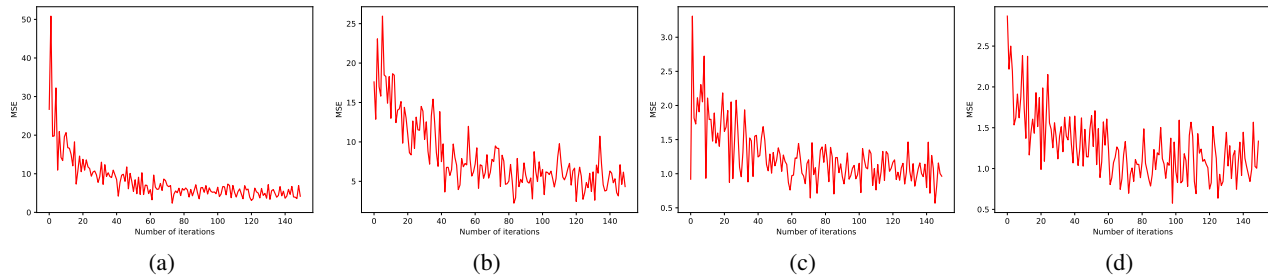


Fig. 10: (a) and (b): Comparison of Non-IID Data and Synthetic Data-Augmented Data for Diabetic. (c) and (d): Comparison of Non-IID Data and Synthetic Data-Augmented Data for Heart Disease data

TABLE V: Contribution of individual components and non-linear applicability(Imp.- Improved, Med. - Medium)

Method	Dia.	Heart	Time	Nonlinear
FL-Normal	5.3176	0.9947	High	Moderate
Synth. Init	Imp.	Imp.	Med.	Improved
Client Balance	5.2776	0.8504	Low	High
Server Distrib.	Imp.	Imp.	Low	High
Full Model	Best	Best	Low.	Highest

full model, which integrates all components, achieves the best overall performance in terms of accuracy, convergence speed, and communication efficiency.

In addition, the nonlinear applicability column indicates that the proposed framework is not restricted to linear models. It provides a qualitative assessment, as the study focuses on isolating the effects of data distribution and synthetic data augmentation independent of model architecture. Linear regression is used as a baseline to ensure controlled evaluation of convergence and communication efficiency under non-IID conditions. Since the framework operates at the data distribution level, it is model-agnostic, and similar improvements are expected when applied to nonlinear models, with the full model demonstrating the highest applicability.

V. RELATED APPROACHES

FL applications span multiple fields ranging from healthcare to finance to the Internet of Things (IoT), etc [14]. However, it is not realistic to assume that data across different participating clients is always IID considering the diverse applications. The existence of non-IID data can result in complications that need to be addressed within the FL framework [5] as discussed below. Various solutions exist for splitting non-IID data among client devices or nodes in FL according to the data partitioning strategies. The techniques encompassed in this category are horizontal [15] and vertical partitioning [16] random sampling [17], clustering-based procedures [18], and methodologies that take into account the inherent features of the data.

FedCD [19] (federated cloning and deletion) is a method for improving federated learning performance in data scenarios without IID.IID. It works by dynamically cloning high-performing models and deleting low-performing ones, enabling devices to self-select into archetypes with similar data distributions.

FedProc [20] is a FL framework that uses global class prototypes to align local training with global objectives, addressing performance drops on non-IID data. Introducing a prototypical contrastive loss and a hybrid network architecture significantly boosts classification accuracy with minimal computational overhead. Variance Reduced Local SGD (VRL-SGD) [21] improves convergence on non-IID data by reducing gradient variance across workers without extra communication cost. Its enhanced version, VRL-SGD-W, adds a warm-up mechanism for greater robustness under high data heterogeneity.

The authors in [22] propose a resource-efficient FL algorithm that addresses non-IID challenges by balancing class distributions and adapting training parameters. It achieves up to 20% higher accuracy with lower computation and communication costs on CIFAR-10 and MNIST. MpFedcon [23] is a personalized FL approach that tackles both label distribution skew and privacy leakage by addressing data heterogeneity and gradient leakage from global and local perspectives. It improves accuracy, privacy protection, and robustness under non-IID data and partial client participation settings.

FedPEC [24] is a high-efficiency hierarchical federated learning (HFL) algorithm that introduces progressive edge collaboration to address non-IID data and communication bottlenecks without relying on unrealistic client allocation. By adaptively adjusting collaborator sets based on similarity and convergence bounds, FedPEC achieves superior accuracy, faster convergence, and lower loss compared to existing FL methods. HyperFed [25] is a FL framework designed to address non-IID challenges by aligning class statistics, leveraging hierarchical information, and ensuring consistent client-server aggregation. It introduces hyperbolic prototypes and aggregation strategies, significantly improving FL performance across multiple datasets.

FLIS [26] is a FL algorithm that tackles non-IID data challenges by clustering clients based on inference similarity, enabling personalized yet collaborative model training within each group. This approach improves performance by aligning clients with similar learning tasks, outperforming state-of-the-

art methods on CIFAR-10/100, SVHN, and FMNIST datasets. Nascimento et al. [27] investigate the impact of different data skew types on FL aggregation algorithms, including FedAvg, FedAvgM, FedProx, FedAdam, and FedYogi, using benchmark and real-world datasets. Their study highlights how attribute, label, quantity, and feature-label skews affect model performance and client heterogeneity. Chen et al. [28] propose FedCure, a semi-asynchronous FL framework designed to mitigate participation bias under non-IID data. FedCure uses coalition construction, participation-aware scheduling, and resource allocation to improve accuracy, efficiency, and long-term scheduling balance. Zeng et al. [29] proposed StoCFL, a stochastically clustered federated learning framework for non-IID data that groups clients with similar data distributions and enables information sharing across clusters, improving model performance and adaptability under dynamic client participation. Singh et al. [30] introduced SelfFed, a self-adaptive federated learning framework that addresses non-IID data and device heterogeneity through adaptive masking, incentive-based aggregation, and reinforcement learning-assisted device selection, improving both accuracy and training efficiency. Li et al. [31] introduced a feature-matching data synthesis method for non-IID FL, demonstrating how synthetic data can reduce heterogeneity and improve model generalization across clients. Table VI compares recent studies on FL non-IID methodologies with the proposed approach. The comparison highlights key characteristics, including experimental settings, non-IID skew types, partition strategies, and supported machine learning models. The proposed framework introduces effective mechanisms to address major challenges associated with non-IID data in federated learning, particularly in terms of data distribution balancing and training efficiency. For the proposed method, five clients were considered, corresponding to five healthcare locations in the Diabetes dataset and five countries in the Heart Disease dataset, thereby simulating a realistic decentralized healthcare FL environment.

A key strength of our algorithm lies in its versatility, capable of handling both linear and non-linear models, thus enhancing its applicability to diverse real-world scenarios with complex data patterns. Our focus on mitigating attribute skew—including data, quantity, and label skew—reflects a deep understanding of distributed data challenges, as unresolved skews can lead to biased models with poor generalization. The integration of synthetic data augmentation demonstrates a proactive effort to enrich training data diversity and mitigate risks, which is crucial in domains like healthcare where data is limited and imbalanced due to privacy constraints. Practical validation is further strengthened through the use of real-world medical datasets on diabetes and heart disease, offering insights that go beyond the limitations of commonly used benchmark datasets. While our method shows superior accuracy, communication efficiency, and faster convergence compared to state-of-the-art solutions, it is important to consider the trade-offs involved—such as the risk of bias from synthetic data and the computational cost of implementation—when applying these improvements in real-

world deployments.

VI. CONCLUSION

Our research successfully addresses the challenge of non-IID data in federated learning, with a focus on horizontally partitioned medical datasets. By enhancing synthetic data strategies to mitigate data skewness and evaluating multiple generation methods, our results demonstrate that the Synthpop technique outperforms others in improving data quality and reliability. Experiments conducted on heart disease and diabetes datasets show significant accuracy improvements, highlighting the effectiveness of the proposed approach. Using linear regression as a proof-of-concept, the framework demonstrates strong potential for handling non-IID medical data and supporting data-driven healthcare applications.

While synthetic data augmentation improves data balance and model convergence, it may introduce potential bias if the generated samples do not accurately reflect the true data distribution. This can lead to distributional shifts or the reinforcement of existing biases, making the quality of synthetic data generation a critical factor. Future work should focus on better quantifying and controlling the fidelity of synthetic samples.

Although synthetic data provides a privacy-preserving alternative to raw data sharing, it may still pose privacy risks if identifiable patterns are retained. Attacks such as membership inference or model inversion could exploit such patterns, highlighting the need for stronger privacy-preserving mechanisms, such as differential privacy.

We also plan to extend our method to vertically partitioned datasets to improve adaptability across diverse data distributions. Future work will further extend the framework to nonlinear models, including neural networks and tree-based methods, to evaluate its effectiveness in more complex real-world scenarios.

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TABLE VI: Comparative Table of FL Non-IID Methodologies. ✓ indicates that the paper refers to the item; ✗ NonIID FL indicates that it does not.

Source	Dataset		Non-IID skew type				Partition			ML Models		
	Clients	Rounds	Attribute skew	Label skew	Quantity skew	Data skew	Not specified	Horizontal	Vertical	Linear	Non-linear	Others
[19]	CIFAR-10	30	✓	✗	✗	✗	✗	✓	✗	✗	✓	✗
[20]	CIFAR10, 100, Tiny-ImageNet	10	✗	✗	✗	✗	✓	✗	✗	✗	✓	✗
[21]	MNIST, DBPedia, Tiny-ImageNet	8	✗	✗	✗	✗	✓	✗	✓	✗	✓	✗
[22]	CIFAR-10, MNIST	100–200	✗	✓	✗	✗	✗	✗	✗	✓	✓	✗
[23]	CIFAR-10, 100, FEMNIST	100, 150	✗	✓	✗	✗	✗	✗	✗	✗	✓	✗
[24]	Fashion-MNIST, CIFAR-10	200	✗	✗	✗	✗	✓	✓	✓	✗	✓	✓
[25]	EMNIST, FMNIST, CIFAR-10, 100	100	✗	✗	✗	✗	✓	✗	✗	✗	✓	✓
[26]	CIFAR-10, 100, SVHN, FMNIST	100	✗	✓	✗	✗	✗	✓	✗	✗	✓	✗
[27]	CIFAR-10, 100, Candidate	50	✓	✓	✓	✓	✗	✗	✗	✗	✓	✗
[28]	MNIST, CIFAR-10, SVHN, CINIC-10	50	✗	✗	✗	✓	✓	✗	✗	✗	✓	✗
[29]	CIFAR10, FEMNIST, Fed-CelebA	200	✓	✓	✗	✓	✗	✓	✗	✗	✓	✗
[30]	KDD99, Credit, Fraud, HAR	200	✓	✓	✓	✗	✗	✓	✗	✗	✓	✗
Proposed	Diabetes, Heart Disease	5	✓	✓	✓	✓	✗	✓	✗	✓	✓	✗

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AVAILABILITY OF CODE

The source code associated with this work is available at:
<https://github.com/NishaTM/NonIID_{FL}>

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